Supplemental Material for: Interlayer electron gas and sodium bistability as the origin of superconductivity in $Na_xCoO_2 \cdot yH_2O$

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This Supplemental Material presents the data, derivations, and consistency checks that the four-page format of the Letter could only summarize, in the order in which the Letter uses them: methods, the well transition, the coupling model, the electronic evidence for the gallery band, the water, the magnetism, the pairing state, the correlation effects not treated explicitly, the design rule and its counterexample, and the predictions that extend beyond this compound.

S1. METHODS

Every number quoted in the Letter derives from one of the calculation sets described here, in sufficient detail that each can be verified or repeated without consulting the code. Table S1 gives the inventory; the rest of this section gives the settings.

Code and functional. All total energies are spin-polarized density-functional calculations in the PBE generalized-gradient approximation, using projector-augmented-wave (PAW) pseudopotentials from pslibrary, run in Quantum ESPRESSO 7.3.1 (GPU build). Plane-wave cutoffs are 60 Ry (wavefunctions) and 480 Ry (density) throughout. Electronic self-consistency uses Marzari–Vanderbilt cold smearing of 0.02 Ry, local Thomas–Fermi charge mixing with mixing_beta = 0.3, and a convergence threshold conv_thr = 10^{-7} Ry for all production total-energy scans; the single exception is the Born–Oppenheimer molecular-dynamics stage (set L, Sec. S7), where conv_thr is loosened to 10^{-6} Ry purely for throughput and every other setting matches set G. Every calculation is spin-polarized (nspin = 2); a direct nspin = 1 vs. nspin = 2 check is reported below, and the local-spin-density over-polarization that separately affects the $\sqrt{3} \times \sqrt{3}$ magnetic moments is discussed in Sec. S10.

Cells. The full-coverage 1×1 cell ($x = 1$, one alkali per formula unit) is used for the main displacement scans, in-plane lattice constants $a = 2.888$ Å (Na), 2.82 Å (Li), and 2.90 Å (K; an ansatz upper estimate bracketing the experimental 2.84–2.87 Å range for K_xCoO_2), with cell areas 7.22, 6.89, and 7.28 Å^2 respectively. The physical $\sqrt{3} \times \sqrt{3}$ cell ($x = 1/3$, $a = 5.00$ Å, area 21.67 Å^2) checks that the transition survives at the real sodium content. Four CoO_2–CoO_2 spacings are computed for each series, $c = 5.5$, 6.9, 8.4, 9.9 Å, bracketing the anhydrous, monolayer-hydrate, and bilayer-hydrate structures actually reported. The alkali sits on the Co-top in-plane site (the lowest-energy registry found in the earlier surface work [1]) at $z = c/2 + \delta$; the oxygen height above the Co plane is held fixed at $z_O = 0.96$ Å for every spacing and every δ, the largest systematic in the calculation (below).

k-meshes. $8 \times 8 \times 3$ for the 1×1 displacement scans, $6 \times 6 \times 3$ for the $\sqrt{3} \times \sqrt{3}$ scans, $6 \times 6 \times 2$ (Gamma-centered) for the explicit-water cell (set G/J/L, Sec. S7), and a dense $24 \times 24 \times 8$ mesh for the non-self-consistent eigenvalues that feed $N(0)$, the density of states, and the Fermi-surface maps of Sec. S4. The band-structure/fatband runs of Sec. S5 follow a 60-point Γ–M–K–Γ path with PROJWFC.X projections onto all Na s/p and Co d orbitals.

Fitting and quantization. $E(\delta)$ is fit to the even Landau polynomial $E_0 + \alpha\delta^2 + \beta\delta^4$ (Eq. (S1)); where the quartic fit is not bracketed by the sampled δ range (the $c \geq 8.4$ Å series, Sec. S2), it is replaced by a constrained sextic, $E_0+\alpha\delta^2+\beta\delta^4+\gamma_6\delta^6$ with $\gamma_6 > 0$ enforced and a hard-core cap $\delta_{\min} \leq c/2 - 2.0$ Å (an ansatz for the alkali–oxygen contact distance; varying it over 1.8–2.5 Å changes only the frozen-regime depths and affects no conclusion). The quantized levels of Sec. S6 come from an exact one-dimensional Schrödinger solve of the fitted well on a parity-resolved, staggered half-line finite-difference grid ($N = 2400$ points, box size following the well), which separates the even and odd sectors exactly and resolves the near-degenerate tunnelling doublet without the numerical noise that a shooting method would introduce. The effective zero-point frequency is always reported as $\hbar\omega_{\text{eff}} = \hbar^2/2M\langle\delta^2\rangle_0$, never as $E_1 - E_0$: in the deep double-well regime $E_1 - E_0$ is an exponentially small tunnelling splitting ($< 10^{-12}$ meV for Na at $c \geq 6.9$ Å), a real feature of the isolated double well but not the scale that couples to electrons, whereas $E_2 - E_0$, the even overtone quantified by $\langle\delta^2\rangle_0$, both sets $\hbar\omega_{\text{eff}}$ and is the boson of Eq. (S3).

Charge and coupling. Bader charges are computed from the converged charge density (valence counts $Z_{\text{val}} = 9$ for Na, 3 for Li, reflecting the semicore PAW pseudopotentials used). The Allen–Dynes evaluation of Sec. S3 uses $\mu^* = 0.10$, is gated on $N(0) \geq 0.1$ states $\text{eV}^{-1}\text{cell}^{-1}\text{spin}^{-1}$ (results are insensitive to this threshold between 0.04 and 0.9), and is cut where λ exceeds the polaron criterion $\lambda_{\text{loc}} = 2$ (surveyed range 1–2 in the literature), beyond which Allen–Dynes theory is not describing a well-defined quasiparticle and raw numbers are not reported.

Water and its relaxation (sets G, J, L). The rigid-cage scans (set G) hold all twelve water atoms fixed while the sodium scans δ (construction in Sec. S7). The adiabatic scans (set J) relax the same twelve water atoms by BFGS at each pinned sodium position, with if_pos freezing every Co, framework-oxygen, and sodium coordinate and leaving all water coordinates free, force convergence threshold forc_conv_thr $= 10^{-3}$ Ry/Bohr and a cap of

100 ionic steps; the sixty-step-capped comparison run in Sec. S7 uses the identical settings, deliberately terminated early. The dispersion-corrected repeat (Sec. S7) adds Grimme D3 (vdw_corr = `'grimme-d3'`, supported natively in QE 7.3.1) on top of the set-J settings. The Born–Oppenheimer molecular-dynamics stage (set L) is generated with only the twelve water atoms mobile (Verlet dynamics, stochastic-velocity-rescaling thermostat, $T = 290$ K, time step ≈ 0.48 fs, 4000 steps ≈ 1.9 ps); it is the open calculation referenced throughout this work and has not yet been analyzed (Table S1, last row).

The frozen-oxygen systematic. The dominant systematic uncertainty in this work arises from holding the CoO_2 oxygen planes fixed at $z_O = 0.96$ Å. A direct check, moving the oxygen plane by ± 0.06 Å (to $z_O = 0.90$ and 1.02 Å at $c = 6.9$ Å, $\delta = 0.5$ Å), shifts the single-point energy by $+117$ and -21 meV respectively, comparable to the well depth itself; every well depth quoted in this work therefore carries a systematic uncertainty of roughly $\pm 50\%$. This uncertainty does not affect the sign of α: the 5.5 Å single well and the 6.9 Å double well survive the full range, and in the frozen polaronic regime the coupling exceeds by an order of magnitude the scale at which a factor of two would be consequential. Two further checks bound the rest of the uncertainty budget. A $12 \times 12 \times 4$ vs. $8 \times 8 \times 3$ k-mesh check at $c = 6.9$ Å, $\delta = 0.5$ Å shifts the total energy by -1.6 meV, negligible on the ~ 100 meV wells. An nspin = 1 vs. nspin = 2 check at the same geometry reproduces the well drop to within 0.2% (-89.1 vs. -88.9 meV); at $c = 9.9$ Å, $\delta = 0.75$ Å the two spin treatments agree to 6.3% (-121.5 vs. -113.8 meV), so magnetism does not affect the well physics at a level that matters for any conclusion in this work. Higher-cutoff (80 Ry) and lower-smearing (0.01 Ry) single points exist at the reference geometry and are consistent with the production settings; the corresponding relative (well-drop) checks are still running and are marked % TODO verify where the manuscript would otherwise imply they are complete. Composition dependence is checked directly rather than assumed: interpolating $\alpha(c)$ for the physical $\sqrt{3} \times \sqrt{3}$ cell gives $c^* \approx 6.57$ Å against 6.40 Å for the full-coverage 1×1 cell, a shift of about 0.2 Å, so the transition is not an artifact of full coverage.

TABLE S1. Calculation inventory. Every displacement scan, relaxation, and dense-mesh run behind the Letter and this Supplement, in the order it is used. "Open" marks the one calculation generated but not yet analyzed.

Set / computes	**Cell**	**Spacings / points**	**k-mesh**	**Feeds**
A: anhydrous $E(\delta)$, Na/Li	1×1	$c = 5.5, 6.9, 8.4, 9.9$ Å; $\delta = 0$–1.00 Å (6 pts)	$8 \times 8 \times 3$	Fig. S1, Fig. 2(a), Table I rows 1,2,4
B: anhydrous $E(\delta)$, Na	$\sqrt{3} \times \sqrt{3}$	$c = 5.5, 6.9, 8.4, 9.9$ Å; $\delta = 0$–0.75 Å (4 pts)	$6 \times 6 \times 3$	Fig. S1, the $\sqrt{3}$ check (Sec. S1)
C: dense NSCF + dos.x, $N(0)$	both cells	same c grid as A/B	$24 \times 24 \times 8$	Fig. 2(b), Fig. S3
D: bands + PROJWFC	1×1, $\sqrt{3} \times \sqrt{3}$	$c = 5.5, 6.9, 9.9$ Å; Γ–M–K–Γ, 60 pts	—	Fig. 3 (Letter), Fig. S4
E: K_xCoO_2 $E(\delta)$	1×1	same c grid as A	$8 \times 8 \times 3$	Fig. S9, c^*_{K}
F: gated $LiCoO_2$ (jellium)	1×1	$c = 5.5$ Å; charge 0.1–0.3 e/f.u.	$8 \times 8 \times 3$	gating prediction, Sec. S12
G: rigid-cage hydrate + vacuum ref.	$\sqrt{3} \times \sqrt{3}$	$c = 9.9$ Å; $\delta = 0$–0.75 Å (4 pts)	$6 \times 6 \times 2$	Fig. 4(a) (Letter)
K: oxygen-height (z_O) scan	$\sqrt{3} \times \sqrt{3}$, vac.	$c = 9.9$ Å; $z_O = 0.90, 1.02$ Å	$6 \times 6 \times 2$	Fig. S10, Δv slope
H: Na_2CoSe_2O check	primitive	SCF + dense NSCF	$24 \times 24 \times 8$	negative control, Sec. S12
I2/J: adiabatic BFGS relax.	$\sqrt{3} \times \sqrt{3}$	$c = 9.9$ Å; $\delta = 0$–1.00 Å (6 pts), 100 steps	$6 \times 6 \times 2$	Fig. 4(a) (Letter)
I3: DFT-D3 spot-check	$\sqrt{3} \times \sqrt{3}$	$c = 9.9$ Å; $\delta = 0.00, 0.15$ converged of 0.00–0.50	$6 \times 6 \times 2$	Sec. S7 D3 check
L/M: BOMD of set-G cell + snapshot rescans	$\sqrt{3} \times \sqrt{3}$	$c = 9.9$ Å; $\delta = 0$, $T = 290$ K, $\sim$1.9 ps	$6 \times 6 \times 2$	*open*: the dynamical test

S2. THE WELL TRANSITION

This section presents the data underlying the value $c^*_{\mathrm{Na}} \approx 6.40$ Å on which the Letter builds: the well shapes at every spacing and both cell sizes, and the check that the transition is not an artifact of full sodium coverage.

For each spacing c the sodium is displaced off centre by δ along c and the total energy $E(\delta)$ recorded; near the bottom the curve is even in δ and is fit by

$$E(\delta) = E_0 + \alpha\,\delta^2 + \beta\,\delta^4, \tag{S1}$$

continued to sixth order where the quartic term alone does not track the data. The sign of α determines the well topology: for $\alpha > 0$ the centre is a stable minimum and the sodium occupies a single well; for $\alpha < 0$ the centre is a local maximum and the well has split in two. Figure S1 shows the full set of curves for both cell sizes. At $c = 5.5$ Å (the dry spacing) $\alpha = +2.5$ eV/Å^2 and the well is single and stiff; by $c = 6.9$ Å (the monolayer-hydrate spacing) $\alpha = -0.5$ eV/Å^2 and the ion sits in a pair of minima about 0.7 Å off centre. The sign flips between two spacings actually computed, so the crossing is bracketed, 5.5 Å $< c^*_{\mathrm{Na}} <$ 6.9 Å, and interpolating the four points places it at $c^*_{\mathrm{Na}} \approx 6.40$ Å. Repeating the full scan at the physical composition $x = 1/3$ in the $\sqrt{3} \times \sqrt{3}$ cell moves the crossing by only about 0.2 Å, so the transition is not an artifact of the full-coverage cell.

Two of the curves in Fig. S1 (open symbols, $c \geq 8.4$ Å) are still falling at the largest displacement sampled, indicating that the sodium is moving toward adsorption onto one CoO_2 sheet rather than settling into a genuine double well; the depths quoted for those spacings are lower bounds, flagged as such in the figure, and this is the pathology that the water-cage construction of Sec. S7 is designed to remove at $c = 9.9$ Å.

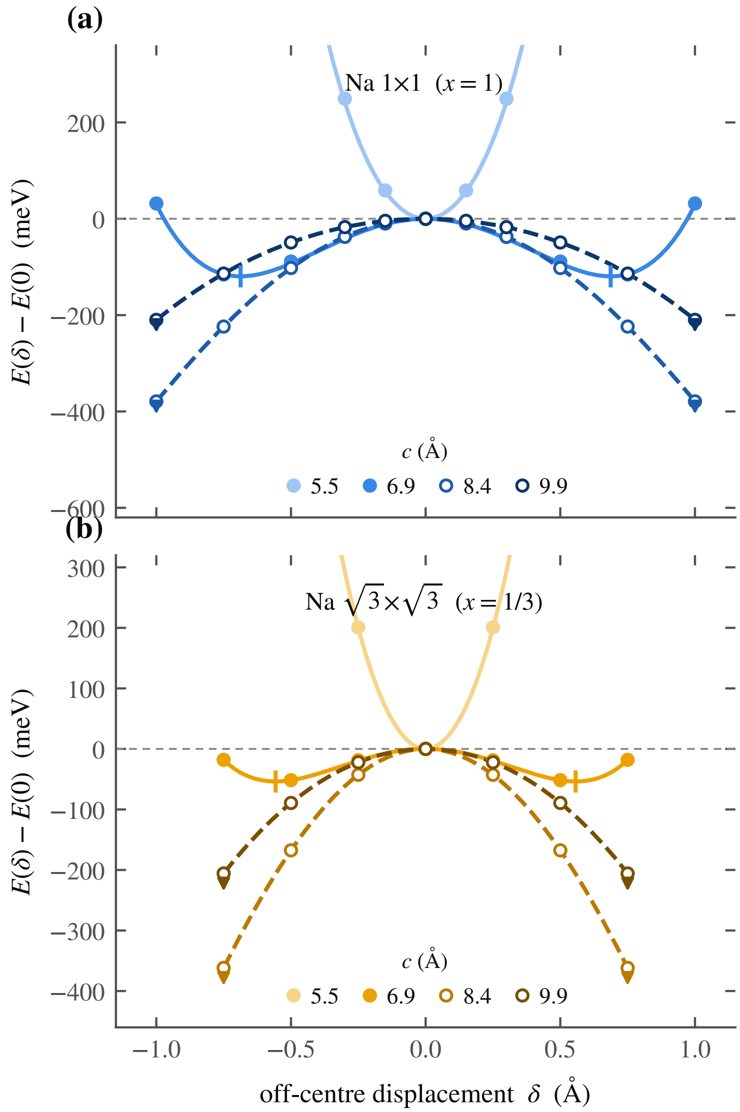

FIG. S1. The single-to-double-well transition. Energy $E(\delta)$ of the off-centre sodium displacement, computed for Na 1×1 (top, $x = 1$) and Na $\sqrt{3} \times \sqrt{3}$ (bottom, $x = 1/3$) at the four gallery spacings; points are total energies, curves the fits of Eq. (S1) continued to sixth order where needed. At 5.5 Å the well is single and stiff ($\alpha > 0$); by 6.9 Å it has split into a double well ($\alpha < 0$) with minima marked. Open symbols with arrows ($c \geq 8.4$ Å) flag scans still falling at the largest sampled displacement, where the sodium is adsorbing onto one layer and the quoted depth is a lower bound only.

S3. THE TWO-BAND MODEL AND THE COUPLING

The Letter states the pairing formula and defers the derivation to this section: the reason the linear coupling vanishes, the form of the quadratic vertex, and the construction of the dome of Fig. 2(c) from $\lambda(c)$ and $\hbar\Omega(c)$.

The bands. The two states that share the gallery electron are taken from an SSH-type four-band tight-binding model developed for the surfaces of the layered cobaltates [1], rather than fitted here; the model is restricted to the pair of bands relevant in the gallery, one of alkali sp_z character and one of CoO_2 character. Their band centres split according to how the electron is shared between them, and that split is fixed by the Bader charge computed directly in Sec. S2, so the gallery band fills exactly as the two-dimensional gas turns on in Fig. 2(b) of the Letter, with no free parameter left in the electronic part of the model. What is added here is the sodium's own motion, and how the two bands respond to it. Figure S2 sketches all three pieces of the construction: the charge-sharing bands (panel a), the parity argument (panel b), and the quantized double well with its overtone boson (panel c).

The parity argument. The sodium displacement δ is odd under reflection through the gallery midplane. A linear electron–phonon coupling would shift a band energy in proportion to δ, and such a term must vanish identically at the symmetric point by that reflection alone; we checked this explicitly in the model at several k_z and it holds everywhere in the gallery band. The leading coupling is therefore quadratic, and the band energy responds as $\frac{1}{2}\chi\delta^2$, with

$$\chi = 2\sum_m \frac{|\langle \mathrm{alk}|\,\partial H/\partial\delta\,|m\rangle|^2}{E_{\mathrm{alk}} - E_m}, \tag{S2}$$

the ordinary second-order perturbation sum over the other bands m of the model. Physically, the sum describes the counter-polarization of the ion's own cloud: the displacement admixes the band of opposite parity, so the gas polarizes against the ionic motion, and the resulting energy gain is even in δ. Near the transition this comes out to $\chi \approx -6$ eV/Å^2, and it grows in magnitude as the band splitting collapses further out in c.

The overtone boson. Because the coupling is quadratic, an electron scatters off two quanta of the sodium mode rather than one. The matrix element that enters is $\langle 0|\delta^2|2\rangle$ between the ground state and the second excited state of the quantized well (coupling to the first excited state is forbidden by the same parity selection rule that eliminates the linear term), and the effective boson is the even overtone $\hbar\Omega = E_2 - E_0$. The dimensionless coupling is

$$\lambda = \frac{2\,N(0)\,g^2}{\hbar\Omega}, \qquad g = \tfrac{1}{2}|\chi|\,\langle 0|\delta^2|2\rangle, \tag{S3}$$

with $N(0)$ taken directly from the density-functional calculation of Sec. S2, with no adjustable parameter.

The dome and its boundaries. Inserting $\lambda(c)$, $\hbar\Omega(c)$, and $N(0)(c)$ into the Allen–Dynes formula, with $\mu^* = 0.10$ and gated on $N(0) \geq 0.1$ states eV^{-1}cell^{-1} spin^{-1}, gives the dome shown in Fig. 2(c) of the Letter: a narrow peak between $c \approx 6.35$ and 6.45 Å, height $T_c \approx 14$ K at $\lambda \approx 1.3$ and $\hbar\Omega \approx 11$ meV. The two boundaries mark the limits beyond which the approximation itself ceases to be reliable. Below c^*, $N(0) \to 0$ and $\lambda \to 0$ with it, because the gallery band is empty of carriers. Above the window, or once the sodium is frozen into one well, $\hbar\Omega$ stays near 20 meV while λ rises past 2, beyond the weak-to-intermediate-coupling regime and into the self-trapped-polaron regime; Allen–Dynes theory does not describe that regime, and a bipolaron treatment past that cut has not been attempted here. The ansatz range quoted for the peak, 11–23 K, comes from varying the electron–phonon range parameter over its plausible values; the dome height depends on that parameter to the fourth power, which is why the Letter states $T_c \approx 14$ K as an order-of-magnitude estimate rather than a precise value.

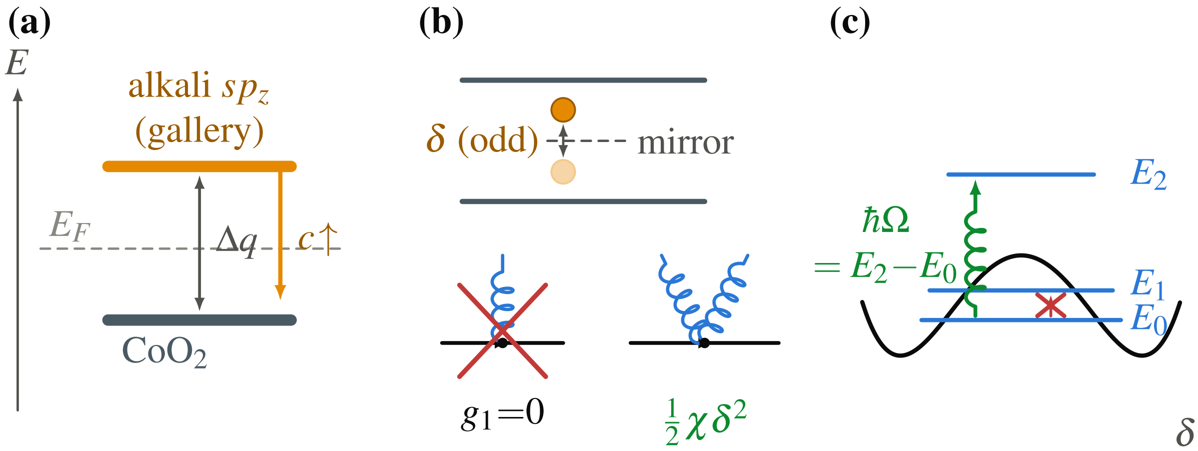

FIG. S2. The two-band model and its coupling. (a) The gallery electron is shared between an alkali sp_z band and a CoO_2 band, taken from the SSH-type surface model [1] and split by the Bader charge computed directly. (b) The gallery midplane mirror forbids a linear electron–phonon vertex at $\delta = 0$; the leading coupling is quadratic. (c) The quantized double well, with the near-degenerate tunnelling doublet E_0, E_1 and the even overtone E_2 that carries the quadratic coupling; the pairing boson is $\hbar\Omega = E_2 - E_0$.

S4. THE FERMI SURFACE AND THE ARPES COMPARISON

The band structure of Sec. S5 shows the gallery band crossing E_F; this section compares the resulting Fermi surface with the photoemission measurement available for comparison.

Figure S3 is the Fermi surface at $c = 9.9$ Å: the spectral function $A(\mathbf{k}, E_F)$ in the $k_z = 0$ plane of the Na 1×1 hexagonal Brillouin zone, from the dense $24\times24\times8$ non-self-consistent eigenvalues, symmetry-completed with the C_{6v} point group and Lorentzian-broadened at E_F (50 meV, both spins summed). The calculation gives a single electron barrel around Γ, the sodium two-dimensional gas, and no e_g' pockets at K. Angle-resolved photoemission on the anhydrous parent $Na_{0.73}CoO_2$ [2] sees the same topology: a single a_{1g} barrel around Γ with measured $k_F \approx 0.65$–0.70 Å^{-1}, and no e_g' pockets, the e_g' band sitting ≈ 85 meV below E_F. Our barrel is sodium- rather than cobalt-derived and the composition differs, so the comparison concerns topology and size rather than band identity; the two barrels are of comparable radius. The measured a_{1g} velocities, $v_F \approx 0.3$ and 0.6 eV Å along Γ–K and Γ–M respectively, lie roughly a factor of two below the band-theory value, a renormalization not

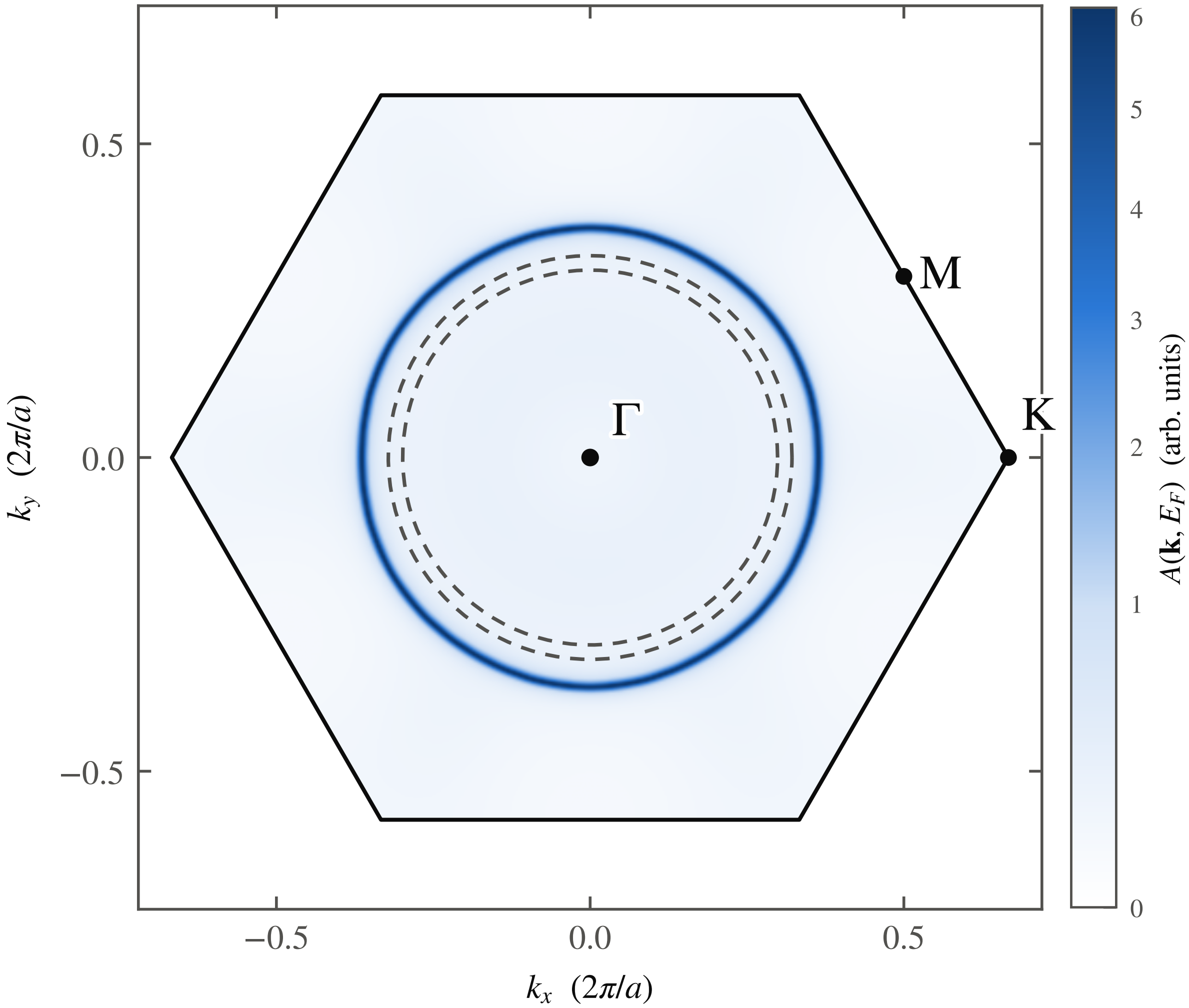

FIG. S3. Fermi surface at $c = 9.9$ Å and the ARPES comparison. $A(\mathbf{k}, E_F)$ in the $k_z = 0$ plane of the Na 1×1 hexagonal Brillouin zone, symmetry-completed and Lorentzian-broadened (50 meV, both spins summed): a single electron barrel around Γ and no e_g' pockets at K. Dashed: the ARPES barrel on anhydrous $Na_{0.73}CoO_2$ [2] ($k_F \approx 0.65$–0.70 Å^{-1}), of comparable radius; the measured a_{1g} velocities are $\sim 2\times$ below the unrenormalized band value shown here.

captured by the self-energy-free Kohn–Sham map (see also Sec. S10).

S5. ELECTRONIC EVIDENCE FOR THE GALLERY BAND

A. At full sodium coverage

Figure S5 (reproduced from the Letter's Fig. 3, with the full caption repeated here for completeness) shows the band structure from which the Letter's central number, 91% sodium weight at E_F by $c = 9.9$ Å, is obtained. It is bare Kohn–Sham spectral weight, both spins summed and each band Lorentzian-broadened by $\Gamma = 40$ meV, with no self-energy applied. The cell is weakly spin-polarized above the transition ($0.8\,\mu_B$ at 9.9 Å), so the gallery band is genuinely spin-split; summing both spins is what a photoemission measurement would also collect.

The Bader charge on the sodium tracks the same filling. In the 1×1 cell it grows monotonically from essentially zero at $c = 5.5$ Å to $0.58\,e$ at 9.9 Å, the number quoted in the Letter. Converted with the in-plane cell area ($a = 2.888$ Å, $A = 7.22$ Å^2), the fully returned charge corresponds to an areal density $n_{2\mathrm{D}} \approx 8 \times 10^{14}$ cm^{-2}; if that charge occupied a single spin-degenerate band, its Fermi radius would be $k_F = \sqrt{2\pi n_{2\mathrm{D}}} \approx 0.71$ Å^{-1}, of the same size as the Fermi-surface barrel computed in Sec. S4 and as the measured barrel on the anhydrous parent ($k_F \approx 0.65$–0.70 Å^{-1}). The Bader count, the Fermi-surface volume, and the factor of ≈ 30 rise of $N(0)$ between 5.5 and 6.9 Å [Fig. 2(b) of the Letter] are three mutually consistent measures of one band filling. The same quantity at the physical composition is smaller: in the $x = 1/3$ cell the sodium Bader charge rises by $\Delta q \approx 0.18\,e$ between the monolayer and bilayer spacings, the value that sets the valence split of Sec. S12.

B. At the physical composition, $x = 1/3$

The full-coverage result is checked at the real sodium content in the $\sqrt{3} \times \sqrt{3}$ cell. Figure S4 mirrors the $x = 1$ case exactly: no sodium weight at E_F at 5.5 Å, a strong sodium band through E_F at 9.9 Å. These are supercell bands, each a fold of the primitive zone; because the plane-wave coefficients required for a weight-resolved unfolding were not retained, we indicate the folding relations rather than construct an unfolded dispersion. The 1×1 K point folds onto the supercell Γ, and the 1×1 M point folds onto the supercell M. Every band in Fig. S4 should be read as folded.

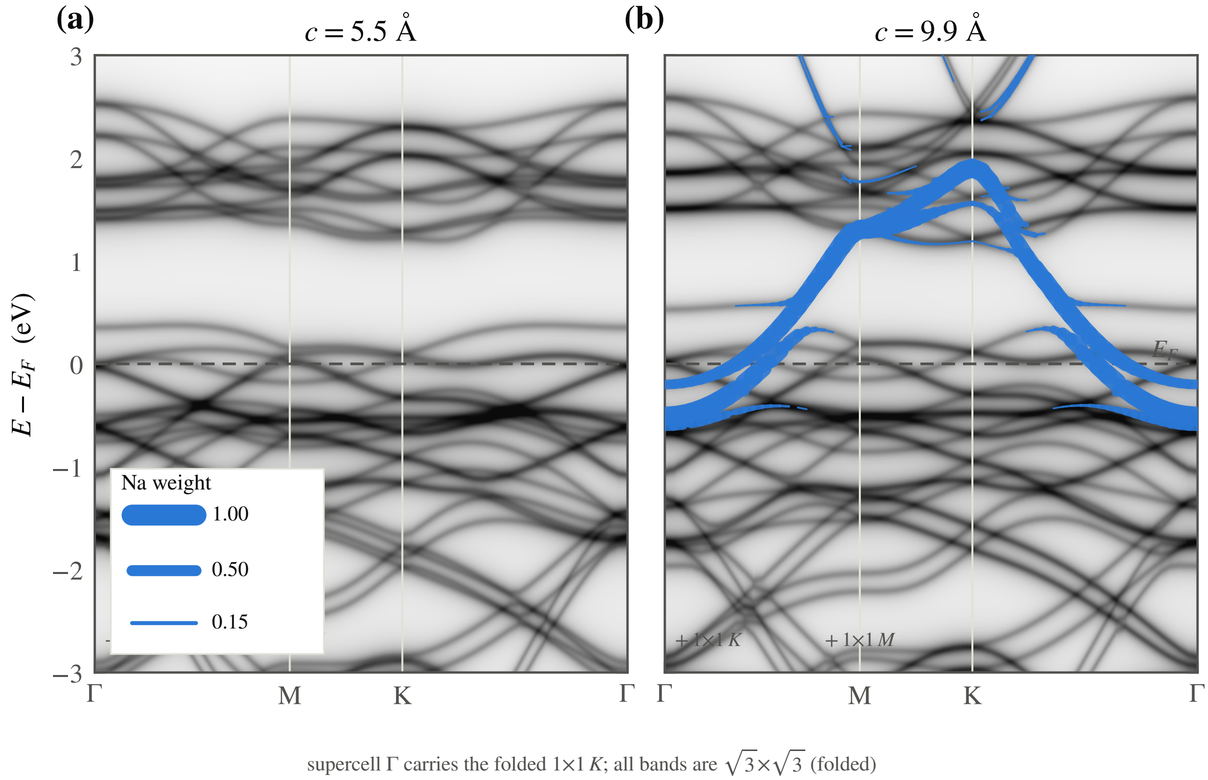

FIG. S4. The gallery band at the physical composition, $x = 1/3$. Kohn–Sham spectral weight of the Na $\sqrt{3} \times \sqrt{3}$ cell at (a) 5.5 and (b) 9.9 Å, built exactly as in Fig. S5 (grayscale total weight, blue sodium fatband, both spins summed, 40 meV broadening). No sodium weight sits at E_F at 5.5 Å; a strong sodium band crosses E_F at 9.9 Å, mirroring the $x = 1$ result. The 1×1 K point folds onto the supercell Γ and the 1×1 M point onto the supercell M; every band shown here is folded.

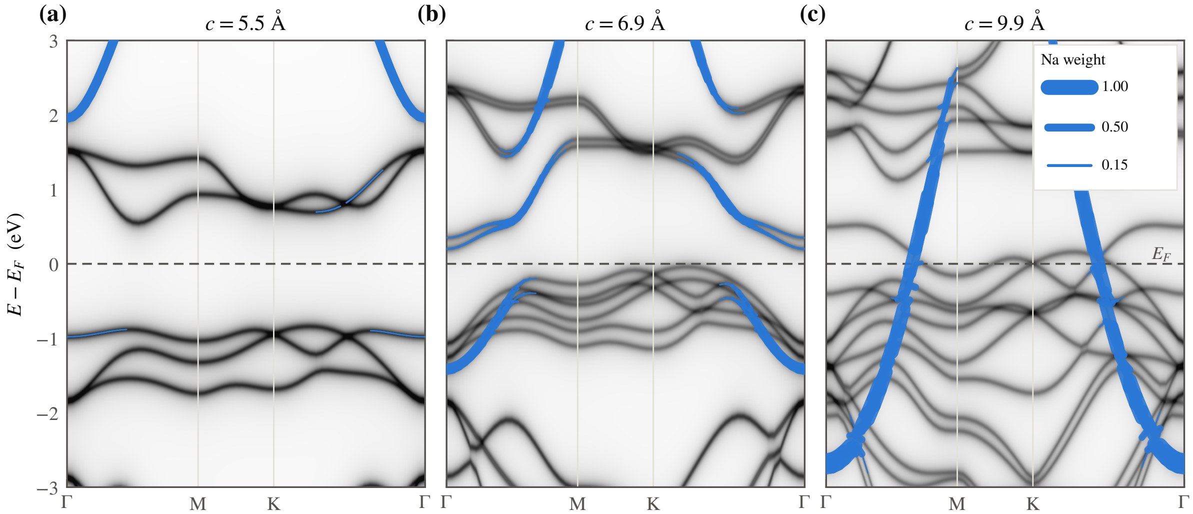

FIG. S5. (Reproduced from the Letter.) Emergence of the gallery band. Kohn–Sham spectral weight of Na 1×1 along Γ–M–K–Γ at (a) 5.5, (b) 6.9, and (c) 9.9 Å, referenced to E_F (dashed). Grayscale is the total DFT spectral weight (all bands, both spins summed, each Lorentzian-broadened by $\Gamma = 40$ meV); the blue fatband is the sodium-orbital projected weight from PROJWFC, its width proportional to sodium character (key in panel c). At 5.5 Å the $x = 1$ cell is a band insulator and the sodium band lies ≈ 2 eV above E_F; by 9.9 Å a band of ∼91% sodium character has descended through E_F, the interlayer electron gas appearing as a single band. No self-energy is applied; this is bare Kohn–Sham weight rather than a renormalized quasiparticle spectrum.

S6. MODE SOFTENING AND THE QUANTIZED WELLS

The pairing model of Sec. S3 requires a soft anharmonic mode, beyond a sign change in α; this section quantifies the softening.

Figure S6 is the effective zero-point frequency $\hbar\omega_{\text{eff}} = \hbar^2/2M\langle\delta^2\rangle_0$ of the quantized alkali mode, obtained from an exact one-dimensional solution of the fitted wells of Sec. S2 at the physical sodium mass. It collapses by three orders of magnitude across the transition, from a stiff ~ 30 meV single-well mode to a sub-meV soft mode as the gallery opens, entering a quantum-paraelectric regime in which the ion is smeared across the barrier rather than localized in either minimum. The inset shows the lowest levels of the 6.9 Å sodium double well directly: the near-degenerate tunnelling doublet E_0, E_1 that the parity argument of Sec. S3 forbids from coupling linearly, and the even overtone E_2 that carries the quadratic vertex and sets $\hbar\Omega$.

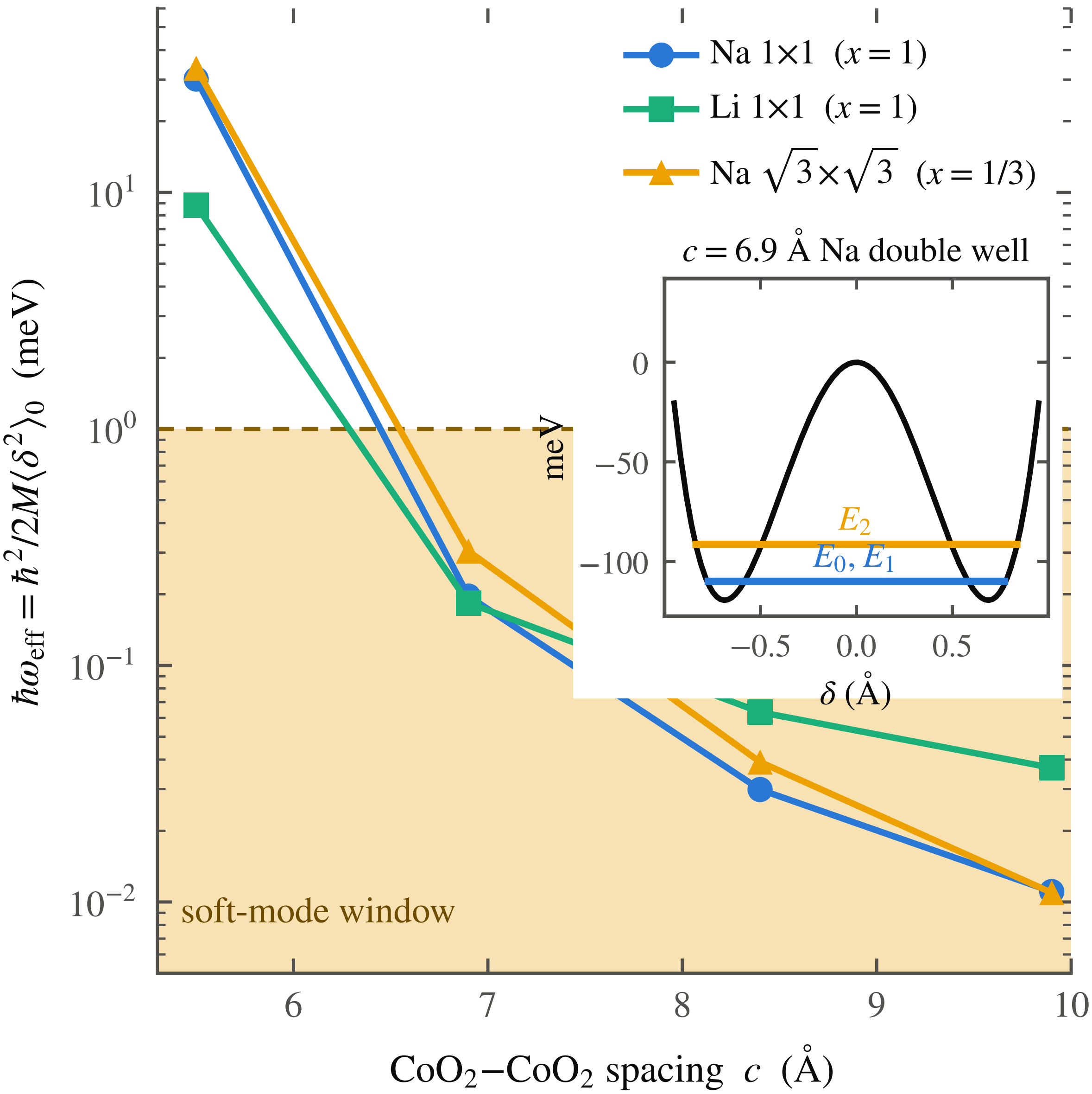

FIG. S6. The sodium mode softening. Effective zero-point frequency $\hbar\omega_{\rm eff} = \hbar^2/2M\langle\delta^2\rangle_0$ of the quantized alkali mode, from an exact one-dimensional solution of the fitted wells at the physical mass. It collapses by three orders of magnitude across the transition, from a stiff ~ 30 meV single-well mode to a sub-meV soft mode, entering a quantum-paraelectric window. Inset: the lowest levels of the 6.9 Å double well, the tunnelling doublet E_0, E_1 and the overtone E_2 that carries the quadratic coupling.

S7. THE WATER CAGE

The Letter's Fig. 4 rests entirely on the construction described here. We describe it in full because the result depends on the construction, and because the three-way comparison (vacuum, rigid cage, adiabatic cage) converts a qualitative scenario into a testable calculation.

A. Construction

The host is the $\sqrt{3} \times \sqrt{3}$ $(x = 1/3)$ Na_xCoO_2 cell at the bilayer spacing $c = 9.9$ Å ($a = 5.00$ Å), the same cell and the same spin-polarized PBE+PAW settings of Sec. S1 used for the anhydrous scans. The CoO_2 sheets and oxygen heights are frozen at the values used throughout, so the water molecules are the only addition. Each water is a rigid gas-phase molecule, O–H bond 0.9572 Å and H–O–H angle 104.52°. Four are placed per cell, in the arrangement neutron diffraction finds for the bilayer hydrate [3, 4]: the sodium sits on the Co-top $(0, 0)$ site on its plane at $z = c/2 + \delta$, and the four water oxygens lie in two layers 1.2 Å below and above that plane, on in-plane sites the sodium does not occupy, two O-top images in the lower layer and two Co-top images in the upper layer. Over the whole displacement scan the sodium–water oxygen distances run 2.05–2.57 Å (lower layer) and 2.92–3.13 Å (upper), an octahedral-like first coordination shell. The hydrogens point away from the sodium, toward the nearest CoO_2 sheet and radially outward, the molecular bisector tilted 50° off the c axis, chosen to maximize the smallest intermolecular contact: every water–water, water–framework, and water–sodium distance stays ≥ 2.05 Å across the whole δ grid. The two down-pointing and two up-pointing molecules cancel the layer's z-dipole exactly, so the cage adds no net dipole field.

B. The rigid cage (set G)

The waters are held fixed while the sodium scans δ, so $E_{\mathrm{hyd}}(\delta) - E_{\mathrm{vac}}(\delta)$, the hydrate curve minus the identical cell with the twelve water atoms deleted, is the clean effect of the solvation cage on the sodium potential, with no relaxation confounds. This probes the static cage only: it determines whether an expanded gallery leaves the sodium chemisorbed, as in vacuum (an unstable mode with $|\hbar\omega| \approx 11i$ meV), or bound in a confined well, as with the cage ($\hbar\omega_{\text{eff}} \approx 23$ meV), and it fixes the frequency of that well. It cannot represent the water's own motion.

C. Adiabatic relaxation (set J)

We next allow the cage to relax. The scan was repeated with the four water molecules relaxed to their minimum-energy configuration at each pinned sodium position, a full BFGS relaxation of the twelve water atoms, one hundred ionic steps, converged at every δ. If the water acted as a lubricant in the naive sense, the relaxing shell would move out of the sodium's path and flatten the well; instead it follows the sodium off-centre and deepens the bound well, from 199 meV (frozen) to 380 meV at the same $\delta \approx 0.30$ Å minimum, and quantized about that minimum the adiabatic mode is stiffer, near 37 meV, rather than softer. Classical adiabatic water therefore does not soften the sodium potential; taken at face value, this result excludes the naive lubricant picture. A dispersion-corrected (DFT-D3) repeat tracks this deepening at the two points where the D3 trajectory has converged, -210 meV against -222 meV (PBE) at $\delta = 0.15$ Å; the remaining D3 points are still relaxing but are consistent with the same trend, so the deepening is not a van-der-Waals artifact of the bare PBE cage.

D. The 0.4 eV configurational spread and the sixty-step artifact

The single relaxed configuration is only the deepest point of a landscape over which the four-water hydrogen-bond network fluctuates. A relaxation of the same $\delta = 0.30$ Å cell, stopped at sixty ionic steps rather than carried to full convergence, comes to rest 394 meV above the fully converged minimum, a different, shallower local minimum of the cage at the identical sodium position (the open star in the Letter's Fig. 4(a)). The hydrogen-bond network therefore presents the sodium with a family of potentials rather than a single one, spanning at least 0.4 eV between local minima at fixed δ; the sodium potential is set by the instantaneous cage configuration, of which the deep adiabatic surface is only the lower envelope. (This also explains why an earlier interpretation of the sixty-step run as a near-flat 5 meV well was an artifact of incomplete relaxation rather than a physical softening; the corrected, fully converged comparison is the one reported throughout.)

E. The timescale argument

The two motions occur on well-separated timescales, and this separation completes the argument. The bound sodium mode has $\hbar\omega \approx 23$ meV, a period near 200 fs; the hydrogen-bond network of gallery water reorganizes on the picosecond scale that quasielastic neutron scattering measures directly [5]. The sodium motion is fast compared with the cage. The ion therefore does not follow the deep adiabatic surface, which the slow network presents only once fully relaxed; it oscillates on a rigid-cage-like surface drawn, at each instant, from the thermal and quantum ensemble of cage configurations, an ensemble that, as the 0.4 eV spread shows, includes surfaces far shallower than the adiabatic minimum. The softening required by this mechanism is therefore statistical-dynamical rather than adiabatic: the fast sodium samples the soft members of the cage ensemble instead of following the slow cage to its stiff, deep minimum. The frozen-cage well is one member of that ensemble and the adiabatic well its extreme deep member; the physical mode lives between them, and the one calculation that can show this directly, a finite-temperature Born–Oppenheimer or path-integral molecular-dynamics sampling of the coupled sodium–water system, remains open.

F. The isotope-null consistency check

This picture makes the deuteration and ^{18}O experiments consistency checks rather than predictions, because both have already been run. Replacing H_2O by D_2O shifts T_c by less than 0.2 K [6]; replacing ^{16}O by ^{18}O is also null [7], with the authors of that study noting explicitly that a null shift does not by itself exclude a phonon mechanism once strong correlations are present. Neither result is evidence against the picture argued here, because neither isotope substitution touches the boson: the pairing mode is the sodium oscillation along c, gated by the statistical ensemble of cage configurations, and the water's own mass enters only through how stiff the hydrogen-bond network is, a second-order effect on the ensemble the sodium samples rather than a first-order mass term in the boson frequency itself. A mechanism that paired electrons directly to a water vibration would, by contrast, predict a first-order isotope shift and is excluded by this result. Sodium itself offers no equivalent test: ^{23}Na is its only stable isotope, so the direct mass test on the boson is unavailable, which is why Sec. S12 emphasizes the spectroscopic search for the mode itself over any further isotope substitution.

S8. SPIN PHASE DIAGRAM AND SPIN-ORBIT COUPLING

The gas that condenses in the gallery is not always paramagnetic, and the location of its magnetic order bears on two long-standing puzzles in this compound: the proximity of a magnetic phase to the superconducting dome, and the coexistence of singlet and triplet assignments in the Knight-shift literature. This section constructs the phase diagram that addresses both.

A. The Stoner strip

Where the gas first condenses, the gallery band is narrow and its density of states is high, the conditions for a Stoner instability. The Radha–Lambrecht surface calculation already finds this polarized gallery gas [1], with the gallery moment antiparallel to the cobalt. Carrying the same SSH-type surface-charge model into the gallery and applying a Stoner criterion to the gallery band gives, in the plane of sodium content x and CoO_2 spacing c, a polarized strip that tracks the carrier turn-on line, giving way to a paramagnon-enhanced metal just outside it and to a conventional paramagnet beyond (Fig. S7(a)); the spacing dependence of the enhancement is the cut in Fig. S7(b). The diagram is anchored to the present calculations: the density-functional moment switches on exactly at the well transition (Fig. S7(c)), which is where the model places the carrier turn-on at full coverage. One quantity is excluded as unreliable: the moment of the $\sqrt{3} \times \sqrt{3}$ cell, over-polarized in the local-spin-density approximation and omitted from the diagram; the well energies on which the mechanism rests shift by less than 6% under the spin treatment (Sec. S10), so the exclusion does not affect the structural conclusions.

In this picture the magnetism is a thin strip adjacent to the carrier turn-on, strongest just past it and weakening as the band widens, so that with increasing spacing the turn-on is followed first by the magnetic strip and then by the superconducting region. This is the ordering Sakurai, Ihara, and Takada report for the magnetic phase bordering the dome against cobalt valence [8]. Figure S8 supports this assignment with experimental data: the measured T_c against cobalt valence, with the superconducting dome and the adjacent magnetic-phase region shown as pastel fills, anchors the adjacency of magnetism and superconductivity in measured data rather than in a schematic. Proximity to the Stoner boundary also resolves a long-standing discrepancy. Early powder NMR suggested a triplet state, whereas later single-crystal work indicated a singlet; in this picture the two results reflect where each sample sits within the strip, because the Stoner enhancement varies steeply across it and hydration drifts from sample to sample, and immediately at the boundary an equal-spin channel of f symmetry lies close in energy to the singlet, connecting to the two f-wave states of Sec. S9 that Mazin and Johannes were unable to eliminate [9]. A specific prediction follows: the NMR Knight-shift suppression below T_c should anticorrelate, across samples, with the normal-state Stoner enhancement. Both symmetry-allowed states of Mazin and Johannes are consistent with the muon null [10], and the chiral $d + id$ scenario [11] is not required to account for the data, though it is not excluded by them.

B. Spin-orbit coupling

Spin-orbit coupling neither selects nor mixes the pairing at any level relevant here. The gallery state is sodium-derived, with only a few percent cobalt weight, so its effective on-band spin-orbit scale is of order 2 meV; the Rashba splitting from the off-centre sodium, estimated from the displaced charge, is about 0.8 meV, of order the gap but far below the Fermi energy, and the parity-mixed component of the gap it induces is at the sub-percent level. On the average centrosymmetric structure the up and down sectors carry opposite Rashba fields and largely cancel. Spin-orbit coupling is therefore a small correction to the spin response and does not enter the pairing.

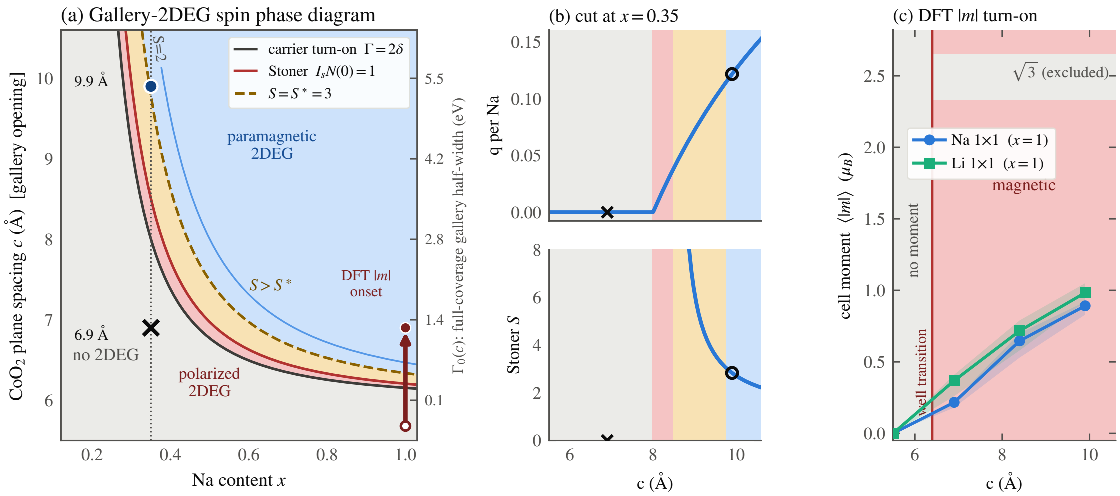

FIG. S7. The gallery-2DEG spin phase diagram. (a) Stoner phase diagram of the interlayer gas, built by carrying the SSH-type surface-charge model of Ref. [1] into the gallery: in the plane of sodium content x and CoO_2 spacing c, the gallery band is empty below the carrier turn-on line, narrow and Stoner-polarized (red) just above it, then a paramagnon-boosted (amber) and plain paramagnetic (blue) metal further out. Markers: the monolayer (6.9 Å, $\times$) and bilayer (9.9 Å, $\circ$) hydrates at the physical $x = 0.35$. (b) Cuts at $x = 0.35$: the carrier fraction q per Na and the Stoner enhancement S against c. (c) The DFT cell-integrated moment $\langle |m| \rangle$ for Na and Li 1×1, switching on at the same spacing the well transition does; the $\sqrt{3} \times \sqrt{3}$ cell is over-polarized in LSDA and shown only as an excluded caveat band. The model is an ansatz calibrated to the one on/off hydrate anchor pair; every parameter is source-tagged in the generating script.

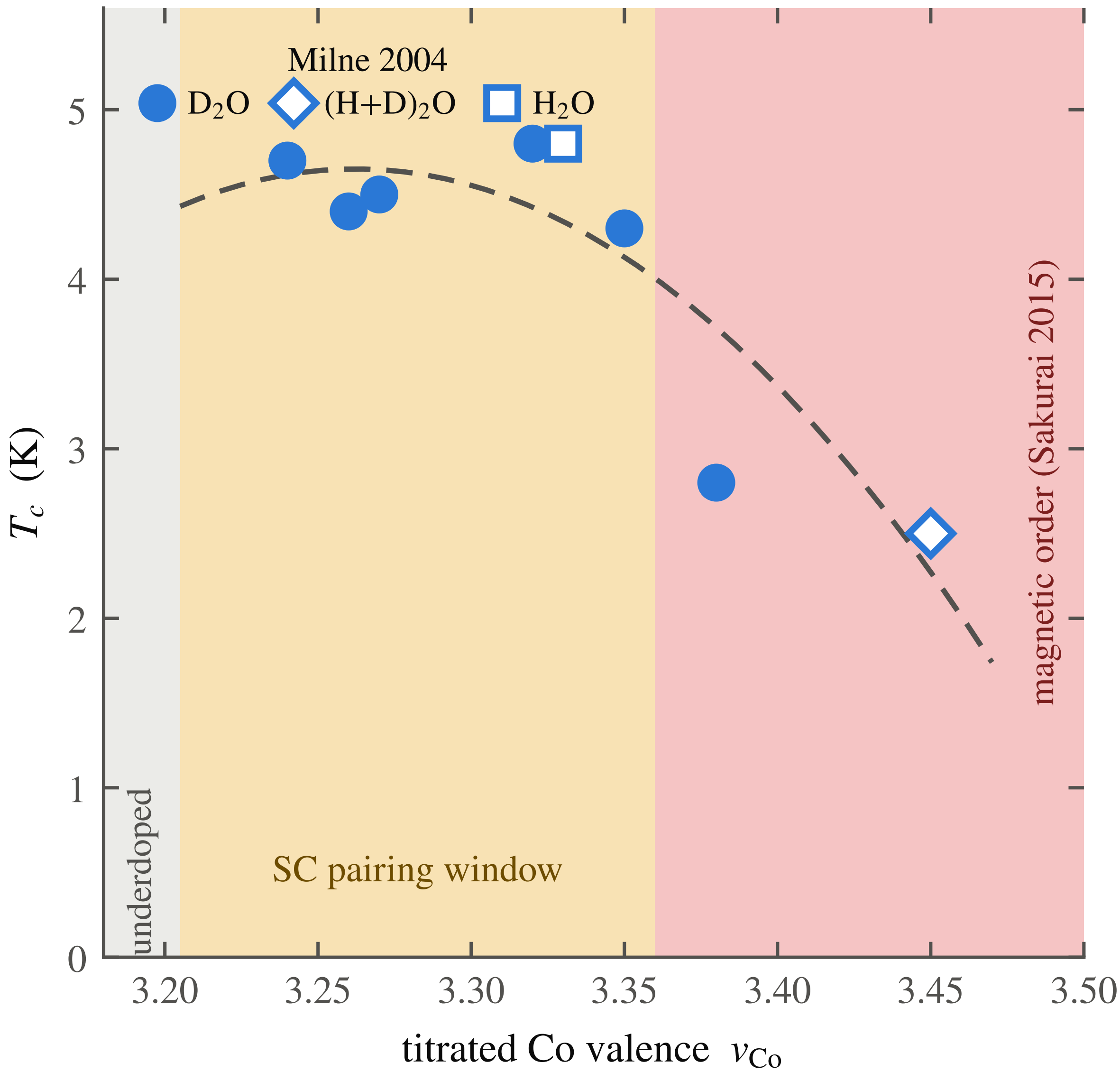

FIG. S8. Experimental evidence that magnetism borders the dome. Experimental T_c against the redox-*titrated* cobalt valence, the eight batches of Milne *et al.* [12], with the superconducting dome and the adjacent magnetic-phase region drawn as pastel fills; the nominal-composition $(4-x)$ points of Ref. [13] use a different, offset (≈ 0.2–0.3) valence axis and are deliberately omitted rather than mixed with the titrated set. The magnetic-phase boundary ($v \approx 3.36$) is anchored to the collapse threshold Milne *et al.* report, combined with the phase adjacency Sakurai *et al.* give [8], and is approximate. The titrated batches cover the optimal plateau and the overdoped collapse but not the underdoped rising edge, so the figure is a partial map of the dome rather than a determination of its full shape; it establishes that the magnetic region sits immediately outside the superconducting one, the adjacency the Stoner strip of Fig. S7(a) predicts from the calculation alone.

S9. EXPERIMENTAL CONSTRAINTS ON THE PAIRING STATE

Proposals for the pairing state of $Na_xCoO_2 \cdot yH_2O$ advanced over two decades are constrained by four key measurements; Table S2 evaluates the principal proposals, including the state advocated in this work, against each of them.

The natural ground state of a phonon-mediated instability on the gallery gas is a spin singlet, anisotropic and $s_\pm$-like across the two bands, the gallery two-dimensional gas and the a_{1g} pocket, and it breaks no time-reversal symmetry. These properties alone satisfy the two most restrictive constraints in Table S2: the singlet is consistent with the single-crystal Knight shift, which excludes the triplet proposals, and the preserved time-reversal symmetry is consistent with the μSR null result [10], which excludes the chiral $d + id$ states [11].

The remaining two constraints are accounted for naturally within the present picture. There is no Hebel–Slichter peak because the pairing boson is a soft, strongly anharmonic mode, and the anharmonic broadening that softens it also smears the coherence peak; we note this as an expected consequence of soft-mode coupling rather than as a fitted result. The specific-heat nodes are the more telling case. The gap on the gallery band is small, set by the dilute gallery density of states, and a small gap on a two-dimensional gas is fragile against precisely the gallery disorder that dehydration introduces. A fragile second gap suppressed below the measurement threshold mimics line nodes, which explains both the node claims and their reported dependence on sample age [8]: the gap carries a fragile second component, removed by the aging of the water network, rather than true nodes. The proposed state is falsifiable in two ways: any observation of broken time-reversal symmetry, a μSR relaxation onset at T_c or a polar Kerr rotation, would exclude the singlet channel, as would a Knight shift that remains constant below T_c in a fresh single crystal, because a singlet requires the shift to decrease. Either observation would exclude the proposed state.

TABLE S2. Experimental constraints on proposed pairing states. Verdicts are single words read against each state's own symmetry: pass, fail, or tension (a constraint met only with an added assumption, flagged [a–f], notes below). The Sakurai entry is a mechanism, not a fixed symmetry, so the symmetry columns do not constrain it. The last row is the state argued in the Letter (Sec. S9).

Proposed state	**Knight shift**	**Hebel–Slichter**	**Heat nodes**	**μSR TRS**
RVB singlet $d+id$ [14–16]	pass	pass	tension[a]	fail
Spin-triplet p/f [17, 18]	fail	pass	pass	pass[b]
Chiral $d+id$, fRG [11]	pass	tension[a]	tension[a]	fail
Two surviving f-wave [9]	tension[c]	pass	pass	pass
s-wave, anharmonic H/O phonons [4]	pass	fail[d]	fail[e]	pass
Magnetic-fluctuation [8]	—	—	—	—
This work: singlet $s_{\pm}$, TRS-preserving, two-gap	pass	pass[f]	pass[g]	pass

[a] Fully gapped states must invoke disorder or a second gap to explain the observed T^2 term. [b] Requires e_g' pockets that ARPES does not see [2]. [c] Passes only after an added reconciliation of the singlet-favoring Knight shift with a triplet order parameter. [d] Predicts a Hebel–Slichter peak the data do not show. [e] Fully gapped; does not reproduce the observed low-T nodal-like term. [f] Anharmonic broadening of the soft-mode boson smears the coherence peak. [g] Small gallery gap is disorder-fragile; explains both the node claims and their sample-age dependence.

S10. CORRELATION EFFECTS BEYOND THE PRESENT TREATMENT

Every estimate above is built on Kohn–Sham bands, and the cobaltates are correlated; this section states the direction in which correlations shift each estimate.

Angle-resolved photoemission on the anhydrous parent finds dispersions narrower than the LDA bands by roughly a factor of two, with a_{1g} velocities $v_F \approx 0.3$–0.6 eV Å against a band value about twice larger [2]. A mass renormalization $m^*/m \approx 2$ raises the density of states $N(0)$ and lowers v_F in the same proportion, and $N(0)$ enters the coupling in the direction that increases it: since $\lambda = 2N(0)g^2/\hbar\Omega$ was built on the unrenormalized density-functional $N(0)$, the λ and T_c estimates of Sec. S3 are, if anything, conservative. The same flattening deepens the narrow-band limit the gallery gas already sits in, and a flatter band at fixed filling carries a higher $N(0)$, which strengthens the Stoner-strip argument of Sec. S8: correlations widen the polarized region of Fig. S7 rather than narrowing it. The natural next step is a DFT+DMFT calculation on the gallery band, which we have not attempted. One caution acts in the opposite direction: in addition to renormalizing the band, correlations can broaden and damp the sodium mode that carries the pairing, and a strongly overdamped mode would couple less efficiently than the sharp Einstein boson assumed here.

The remaining approximations are summarized briefly. The transition spacing is computed at full sodium coverage, $x = 1$, and checked at the physical $x = 1/3$ in a single cell, where it moves by about 0.2 Å. The local-spin-density treatment over-polarizes the $\sqrt{3}$ cell, giving an unphysically large and flat moment there; we base the magnetic analysis on the 1×1 series and exclude the $\sqrt{3}$ moments, and the well energies themselves are insensitive to the spin treatment at the few-percent level, so the over-polarization does not affect the wells. The explicit water (Sec. S7) is treated statically, a rigid cage frozen while the sodium scans and an adiabatically relaxed cage at each sodium position; together they settle the static question and show that adiabatic relaxation deepens rather than softens that well, but neither can represent the statistical-dynamical softening the moving network supplies, the one calculation left open. The pairing is treated as a single Einstein boson in the Allen–Dynes approximation, with an explicit cut where the coupling leaves the regime that approximation can describe; a real bipolaron treatment past that cut is not attempted.

S11. THE DESIGN RULE, PRIOR ART, AND THE LIZRNCL NULL HYPOTHESIS

If the picture holds, it constitutes a design rule rather than an account of a single compound; this section states the rule, identifies the ingredients with precedent in the literature, and tests the rule against the one layered intercalate that appears, at first inspection, to violate it.

Each ingredient of the rule has precedent. Pairing tuned to a soft polar mode at a quantum-critical point is established in $SrTiO_3$ [19–22]; an occupied interlayer state acting as the switch for superconductivity in an intercalate, controlled by the layer separation, is established for the graphite intercalation compounds [23, 24]; enhancement of T_c as an intercalant mode softens, followed by collapse when the ion orders, is observed in CaC_6 under pressure [25]; a rattling alkali ion in an oversized cage driving strong-coupling pairing is realized in the β-pyrochlores [26, 27]; and carriers in one layer paired by an adjacent quantum-paraelectric layer were proposed by Shenoy, Subrahmanyam, and Bishop [28]. What is new here is the combination: the donor ion is itself the quantum-paraelectric unit rather than a caged spectator or a host-lattice mode, and a single structural parameter, the gallery spacing, simultaneously switches on the carriers and tunes the ion through its bistability, so the window closes on both sides rather than one. Stated as a rule: when a layered gallery is opened past the spacing at which the donor ion becomes bistable, the same structural change that transfers the ion's electrons into an interlayer gas softens the ion into the pairing boson.

A rule stated this broadly must be tested against counterexamples, and the most extensively studied layered donor intercalate provides one. In Li_xZrNCl and its hafnium analogue [29], opening the gallery from 9 to more than 22 Å does not suppress the superconductivity: T_c rises modestly and then saturates (Fig. S9(a)), and it survives with no intercalant ion at all, whether the chlorine is de-intercalated or the carriers are supplied by a gate [30]. This would constitute a refutation only if the premise of the rule applied, and it does not: here the donor's electron does not return to the gallery. It fills an in-plane Zr/Hf d–N p host band, the ion remains fully ionized with no spectral weight at the Fermi level, and the pairing belongs to the host rather than to any gallery gas or soft ion. The nitride therefore lies outside the scope of the rule; it tests a different pairing channel, and it constrains the present proposal in two ways. First, it excludes a universal statement of the rule: the rule holds only where the interlayer state lies at the Fermi level, the precondition made computable by Csányi and co-workers [23], and fails where the host band lies below it. Second, it fixes the null hypothesis: a gallery opened without a soft donor ion gives a monotonic, saturating $T_c(c)$, and the nonmonotonic dome computed for the hydrate (Fig. S9(b)) is, against that background, the anomaly that requires the additional ingredient, the soft sodium ion, absent in the nitride.

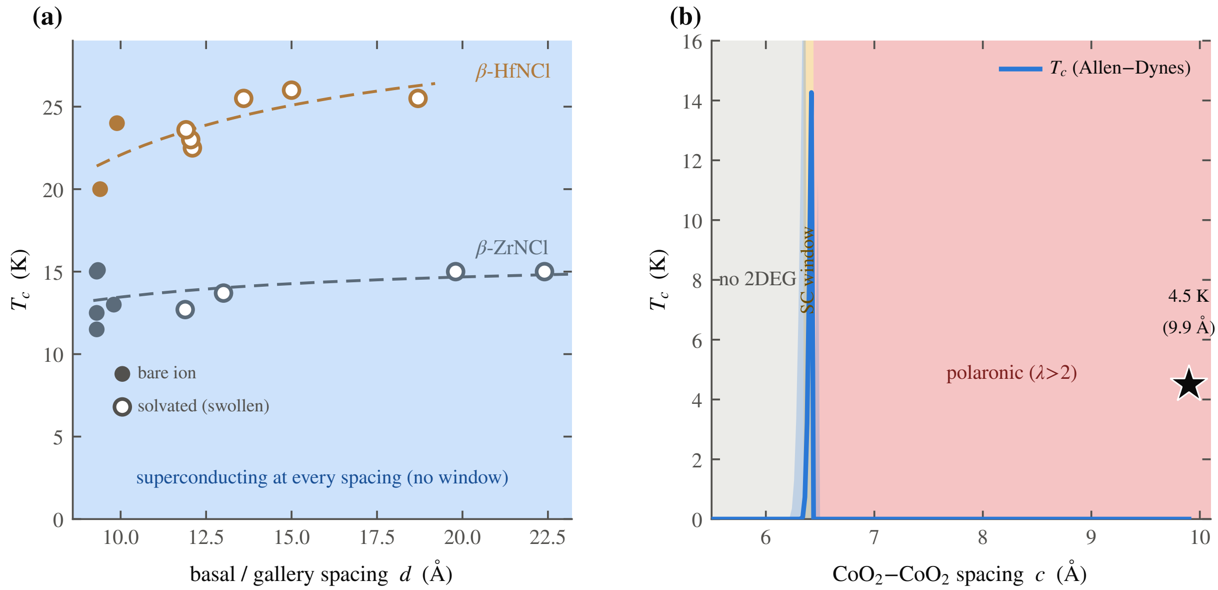

FIG. S9. Test of the null hypothesis. (a) Literature T_c against gallery spacing for Li_xZrNCl and its hafnium analogue [29, 30] (17 points spanning both chemical and gate-induced intercalation): opening the gallery from 9 Å to beyond 22 Å gives a monotonic, saturating $T_c(c)$, because the donor electron never returns to the gallery. (b) Our computed Allen–Dynes dome for the hydrate, with the 4.5 K bilayer-hydrate anchor (star), rises and falls instead: a nonmonotonic window that appears only when the gallery hosts a soft donor ion. The comparison between (a) and (b) defines the null hypothesis against which the present mechanism is tested.

S12. PREDICTIONS BEYOND THE COMPOUND

If the rule of Sec. S11 is correct, it carries implications for neighboring compounds and for probes not yet applied to this one; this section collects those predictions and quantifies them where possible.

A. Lithium, sodium, and potassium

Lithium cobaltate shares the structure with a smaller ion, and superconductivity has never been observed in it; the present picture requires this. Its well crosses into the double-well regime only at the smallest spacing in the dataset, so there is only an upper bound $c^*_{\mathrm{Li}} \leq 5.5$ Å, and at that spacing the well is only 4 meV deep while the lithium zero-point energy is 8.8 meV, larger than the well itself. The light ion does not localize in either minimum; it is smeared across both, a quantum paraelectric, dynamically centred despite the negative α, and a dynamically centred ion provides no soft anharmonic mode. Lithium therefore fails to supply the pairing boson because of its low mass. A heavier gallery ion shifts the threshold in the opposite direction: for potassium the Landau coefficient crosses zero at $c^*_{\mathrm{K}} \approx 6.75$ Å, against 6.40 Å for sodium and ≤ 5.5 Å for lithium, ordering Li $<$ Na $<$ K with ionic size (Fig. S10); hydrated to that spacing, potassium cobaltate should superconduct by the same route with a heavier, softer mode, a direct prediction for hydrated $\mathrm{K}_x\mathrm{CoO}_2$.

B. The valence signature

The gallery gas produces a signature in the sodium valence that two classes of probe should measure differently. If the sodium keeps a fraction q of its electron in the gallery rather than giving it to the cobalt, a site-specific valence probe, the cobalt L-edge or NQR, and a charge-counting probe, redox titration of the whole crystal, should disagree, and only once the gallery is open. This can be quantified: the Bader charge on the sodium in the $x = 1/3$ cell rises from the empty-gallery monolayer spacing to the superconducting bilayer by $\Delta q \approx 0.18\, e$, so the gallery-specific valence split is $\Delta v = x\, \Delta q \approx 0.06$ per cobalt, switching on with the water and vanishing in the monolayer hydrate where the gallery band is empty. This small splitting is distinct from the large valence offset reported in these materials [12], which requires an additional electron reservoir such as oxonium and is several times larger; the present value is the small, gallery-specific contribution alone. Figure S11 gives the microscopic slope behind the number: the Landau coefficient α against the frozen oxygen height z_O at $c = 9.9$ Å (vacuum cell, three collinear points: the production $z_O = 0.96$ Å and two dedicated checks at $z_O = 0.90$ and 1.02 Å) has slope $d\alpha/dz_O \approx -0.52$ eV/Å^2 per Å, the same charge-transfer channel that sets Δv, made explicit as a single derivative rather than left implicit in the total-energy scan. Only the slope is defensible here, since three points determine a line but not an absolute stiffness; it is presented as the microscopic rate underlying Δv rather than as a fourth independent calibration of it.

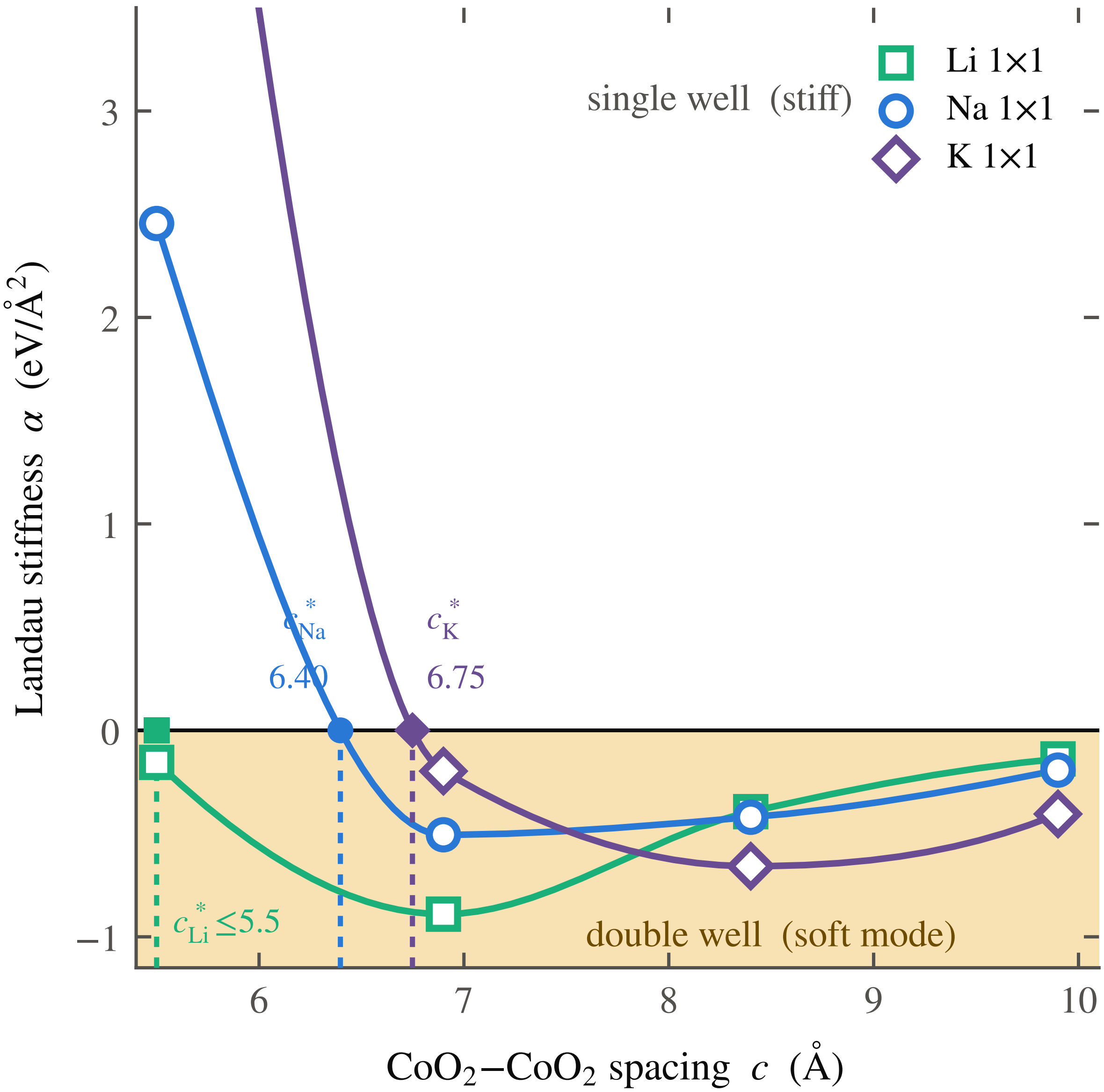

FIG. S10. The threshold ordering. $\alpha(c)$ for Li, Na, and K 1×1, with the three thresholds marked: $c^*_{\mathrm{Li}} \leq 5.5$ Å (an upper bound only, α is already negative at the smallest spacing sampled, not a bracketed crossing), $c^*_{\mathrm{Na}} \approx 6.40$ Å, and $c^*_{\mathrm{K}} \approx 6.75$ Å (both bracketed crossings). The threshold rises with ionic size, the prediction behind the hydrated-K_xCoO_2 test proposed in the Letter.

C. Gating and a negative control

A $LiCoO_2$ thin film in which the carrier density is set electrostatically rather than chemically should show the gallery turn-on without the disadvantage of the light frozen ion.

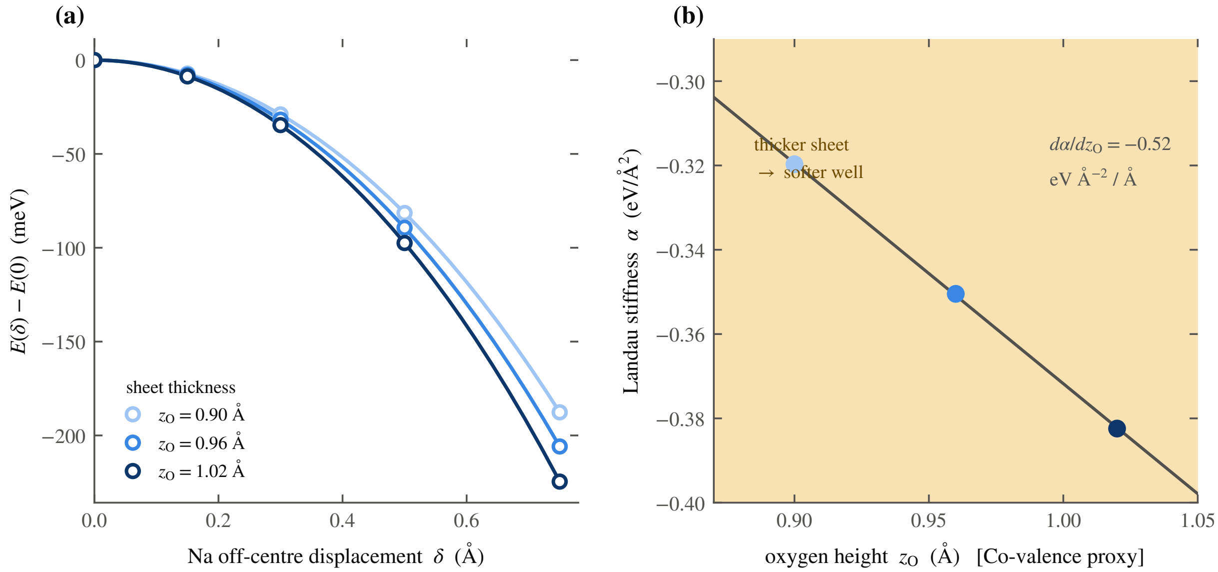

FIG. S11. The microscopic basis of the valence prediction, all vacuum (water-free) potentials at $c = 9.9$ Å. (a) The raw $E(\delta)$ curves at the three sampled oxygen heights $z_O = 0.90$, 0.96 (production), and 1.02 Å, shown for transparency. (b) The extracted Landau coefficient $\alpha(z_O)$: three collinear points give slope $d\alpha/dz_O \approx -0.52$ eV/Å^2 per Å. The defensible quantity is this slope, not an absolute well stiffness; it is the same oxygen-to-gallery charge-transfer channel that sets the valence split $\Delta v \approx 0.06$ per cobalt quoted in the text.

Adding carriers to the anhydrous (5.5 Å) lithium cell by a compensating background, 0.1–0.3 e per formula unit, fills the gallery: the Fermi level and the areal density rise monotonically, to $n_{2D} \approx 4 \times 10^{14}$ cm^{-2} at $0.3\,e$, and the lithium well softens as it fills while remaining under a millielectronvolt deep, far below the lithium zero-point energy of ~ 9 meV, so the ion stays dynamically centred. Beyond its role as a prediction, this calculation serves as a control for the mechanism itself: the charge is added with the lattice fixed at the closed spacing, and the gallery band fills and the well softens nonetheless, demonstrating that the carrier return, and not the expansion per se, is the agent of both changes (this is the control cited in the Letter). Gating thus tunes the gas without freezing the ion, allowing the gas and the magnetism to be tested independently of the pairing boson; this is a prediction for ionic-liquid-gated thin films.

Na_2CoSe_2O [31], a mixed-anion cobaltate ($T_c = 6.3$ K) that carries the same two-dimensional cobalt network with a different gallery chemistry, provides a test of whether its gallery ion hosts an interlayer band. It does not: the states at the Fermi level are the Co $3d$–Se $4p$ network, and the ionic sodium of its Na_2O gallery sits well above the Fermi level, with no sodium-derived band there. Na_2CoSe_2O superconducts through its cobalt network rather than through the gallery, and serves as a negative control: the same CoO_2-like sheet, with no gallery gas, superconducting by a different route.

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